STAFF REVIEW · H446 216 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. The largest value in the unsorted section.

2. A condition that stops recursive calls.

3. 4; indexes begin at 0.

4. True, because 4 is less than or equal to 6.

I Do — source worked steps

1. Predict the next state — 6 is larger than 5; 3 is smaller and swaps into the left boundary.

2. Reveal and explain — 8 is larger; 2 swaps to index 1, giving [3,2,8,6,5].

3. Reveal and explain — Swap pivot 5 with position 2: [3,2,5,6,8]. The two sides are not yet necessarily sorted.

We Do — guided CHECK

1. Reveal and explain
7 is too large, then 3 moves to index 1: [1,3,7,8,6].
Put pivot at index 2: [1,3,6,8,7]. The right side [8,7] still needs sorting.

You Do — source mark schemes

1. [3] Pivot=3 [1]; move 1 left [1]; result [1,3,4] with pivot index 1 [1].

2. [5] Move 2: [2,9,5,1,6] [1]; move 5: [2,5,9,1,6] [1]; move 1: [2,5,1,9,6] [1]; pivot swap: [2,5,1,6,9] [1]; pivot index 3 [1].

3. [4] An already sorted input can place each pivot at one extreme [1]; partitions become n−1 and 0 elements [1]; recursion depth can become O(n), while total comparisons become $O(n^2)$ [1]; randomised/median-of-three pivot selection can reduce repeated poor pivots in practice [1].

Exit Ticket — source answer

1. False. Partition places the pivot correctly; the two sides require further sorting.